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Dashboard for multi site management system

Abstract

A multi-site Building Management System (BMS) monitors performance of a local BMS at each of a plurality of remote sites. The multi-site BMS includes a controller that is configured to determine a plurality of local performance metrics associated with each local BMS based on the operational data received from each local BMS and to aggregate like ones of the plurality of local performance metrics, resulting in a plurality of aggregated performance metrics. The controller is configured to display on the display a plurality of panels, to display in each of the plurality of panels the corresponding one of the plurality of aggregated performance metrics and to display in each of the plurality of panels a ranking of one or more of the remote sites by their corresponding local performance metric.

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Background/Summary

(1) This is a continuation of co-pending U.S. patent application Ser. No. 17/345,955, filed Jun. 11, 2021, and entitled “DASHBOARD FOR MULTI SITE MANAGEMENT SYSTEM”, which claims the benefit of U.S. Provisional Application No. 63/039,373, filed Jun. 15, 2020, both of which are hereby incorporated by reference.

TECHNICAL FIELD
(1) The present disclosure relates generally to building management systems, and more particularly to multi-site building management systems.
BACKGROUND
(2) Portfolio managers may be responsible for monitoring tens, hundreds or even thousands of different building locations that may be spread out across different states or even across different nations. Each of the building locations may have a local building management system that provides data on alarms, energy conservation and the like. It can be difficult to easily spot potential problems occurring at a single building location, much less from a multitude of building management systems that are spread out geographically. It will be appreciated that the sheer volume of data, even if limited for example to active alarms, can be overwhelming. What would be desirable would be a multi-site management system that can help a portfolio manager manage the data coming in from a number of different building management systems.
SUMMARY
(3) The present disclosure relates generally to helping a portfolio manager manage the sheer volume of data coming in from a number of different building management systems and assist the portfolio manager in quickly and efficiently detecting and responding to potential issues throughout the portfolio of buildings for which they are responsible. In an example, a multi-site Building Management System (BMS) monitors performance of a local BMS at each of a plurality of remote sites. This example multi-site BMS includes a

port, a display and a controller that is operatively coupled to the display and the port. The port receives operational data from the local BMS of each of the plurality of remote sites. The controller is configured to determine a plurality of local performance metrics associated with the local BMS of each of the plurality of remote sites based on the operational data received from the local BMS of each of the plurality of remote sites. The controller is further configured to aggregate like ones of the plurality of local performance metrics from the plurality of remote sites, resulting in a plurality of aggregated performance metrics. The controller is further configured to display on the display a plurality of panels, each panel associated with a different one of the plurality of local performance metrics. The controller also displays in each of the plurality of panels the corresponding one of the plurality of aggregated performance metrics. The controller also displays in each of the plurality of panels a ranking of one or more of the remote sites by their corresponding local performance metric, sometimes with outliers ranked first so they are easily identified and accessed.
(4) In another example, a non-transient computer readable medium has instructions stored thereon. When the instructions are executed by a processor, the processor is caused to determine a plurality of local performance metrics associated with a local BMS of each of a plurality of remote sites based on operational data received from the local BMS of each of the plurality of remote sites. The processor is further caused to aggregate like ones of the plurality of local performance metrics from the plurality of remote sites, resulting in a plurality of aggregated performance metrics. The processor is further caused to display on the display a plurality of panels, each panel associated with a different one of the plurality of local performance metrics. The processor is further caused to display in each of the plurality of panels the corresponding one of the plurality of aggregated performance metrics. The processor is also caused to allow a user to select one of the plurality of remote sites, and in response to selection of one of the plurality of remote sites, display a site view that includes at least some of the local performance metrics associated with the particular selected remote site.
(5) In another example, a method monitors a performance of a local BMS at each of a plurality of remote sites. A plurality of local performance metrics associated with a local BMS of each of a plurality of remote sites are determined based on operational data received from the local BMS of each of the plurality of remote sites. Like ones of the plurality of local performance metrics from the plurality of remote sites are aggregated, resulting in a plurality of aggregated performance metrics. A plurality of panels are displayed on a display, each panel associated with a different one of the plurality of local performance metrics. The corresponding one of the plurality of aggregated performance metrics are displayed in each of the plurality of panels. A ranking of one or more of the remote sites by their corresponding local performance metric is also displayed in each of the plurality of panels.
(6) The preceding summary is provided to facilitate an understanding of some of the innovative features unique to the present disclosure and is not intended to be a full description. A full appreciation of the disclosure can be gained by taking the entire specification, claims, figures, and abstract as a whole.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) The disclosure may be more completely understood in consideration of the following description of various examples in connection with the accompanying drawings, in which:
(2) FIG. **1** is a schematic block diagram of an illustrative multi-site BMS operatively coupled to a number of remote sites;
(3) FIG. **2** is a schematic block diagram of an illustrative multi-site BMS usable in the illustrative building system of FIG. **1**;
(4) FIG. **3** is a flow diagram showing an illustrative method using the illustrative multi-site BMS of FIG. **2**;
(5) FIG. **4** is a flow diagram showing an illustrative method using the illustrative multi-site BMS of FIG. **2**;
(6) FIG. **5** is a flow diagram showing an illustrative method using the illustrative multi-site BMS of FIG. **2**; and
(7) FIGS. **6** through **11** are illustrative dashboard screens that may be generated by the illustrative multi-site BMS of FIG. **2**.
(8) While the disclosure is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the disclosure to the particular examples described. On the

contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.

DESCRIPTION

(9) The following description should be read with reference to the drawings, in which like elements in different drawings are numbered in like fashion. The drawings, which are not necessarily to scale, depict examples that are not intended to limit the scope of the disclosure. Although examples are illustrated for the various elements, those skilled in the art will recognize that many of the examples provided have suitable alternatives that may be utilized.

(10) All numbers are herein assumed to be modified by the term “about”, unless the content clearly dictates otherwise. The recitation of numerical ranges by endpoints includes all numbers subsumed within that range (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).

(11) As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include the plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

(12) It is noted that references in the specification to “an embodiment”, “some embodiments”, “other embodiments”, etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is contemplated that the feature, structure, or characteristic is described in connection with an embodiment, it is contemplated that the feature, structure, or characteristic may be applied to other embodiments whether or not explicitly described unless clearly stated to the contrary.

(13) FIG. **1** is a schematic block diagram of an illustrative building management system **10**. In its broadest terms, the illustrative building management system **10** includes a multi-site BMS **12** and a plurality of remote sites **14** operatively coupled to the multi-site BMS. While a total of three remote sites **14** are shown, it will be appreciated that this is merely illustrative, as the multi-site BMS **12** may oversee and/or monitor operations of a large number of remote sites **14**. The remote sites **14** may be distributed across a large geographic area. Each of the remote sites **14** are individually labeled as **14***a*, **14***b*, **14***c* and may each represent any of a variety of different types of sites. While each of the remote sites **14** may be described herein as being buildings, this is not required in all cases. For example, some of the remote sites **14** may also represent factories or other processing facilities.

(14) In the example shown, each of the remote sites **14** include a local BMS **16**, individually labeled as **16***a*, **16***b*, **16***c*. In some cases, some of the remote sites **14** may not include a local BMS **16**. In such cases, the equipment **18**, **20** and/or controllers (not illustrated) that control operation of the equipment **18**, **20** may communicate directly with the gateway **22**. In some cases, information pertaining to operation of the equipment **18**, **20** may be accessible by logging into a local system (not illustrated), or even into the local controllers, with a local dashboard displayed on a web browser or a smart device such is a tablet or smart phone.

(15) Each local BMS **16** may be considered as being operably coupled with a variety of different equipment **18**, **20** that is located at the remote site **14**. It will be appreciated that there will typically be many more pieces of equipment **18**, **20** than the two that are illustrated at each remote site **14**. The equipment **18**, **20** is individually labeled as **18***a*, **20***a*, **18***b*, **20***b*, **18***c*, **20***c*, and may include Heating, Ventilating and Air Conditioning (HVAC) system components. The equipment **18**, **20** may include lighting system components, security system components, and the like. Each BMS **16** may be configured to receive operational data from the equipment **18**, **20** and to formulate control commands for the equipment **18**, **20** in response to the received operational data. Each local BMS **16** may be configured to enable local control of the equipment **18**, **20**, as well as local monitoring of the equipment **18**, **20**.

(16) In some cases, the local BMS **16** may be configured to provide operational data to the multi-site BMS **12**. In order to communicate with the multi-site BMS **12**, in some cases each of the remote sites **14** may include a gateway **22**, individually labeled as **22***a*, **22***b*, **22***c*. The gateways **22**, if present, may provide a way by which each local BMS **16** can communicate with the multi-site BMS **12**. The gateways **22** may provide a means for operational data to be uploaded from each local BMS **16** to the multi-site BMS **12** as well as control commands to be downloaded from the multi-site BMS **12** to each local BMS **16**. In some cases, the gateways **22** may be configured to download software packages from the multi-site BMS **12** that better

configures each local BMS **16** for communication with the multi-site BMS **12**.
(17) FIG. **2** is a schematic block diagram of the illustrative multi-site BMS **12**. The multi-site BMS **12** may be considered as being configured to monitor the performance of the local BMS **16** at each of the remote sites **14**. The multi-site BMS **12** includes a port **24** that is configured to receive operational data from the local BMS **16** at each of the remote sites **14**. The multi-site BMS includes a display **26** and a controller **28** that is operatively coupled to the port **24** and to the display **26**. The controller **28** may be configured to determine a plurality of local performance metrics associated with the local BMS **16** of each of the plurality of remote sites **14** based on the operational data received from the local BMS **16** of each of the plurality of remote sites **14**. One of the local performance metrics may be associated with alarms that are issued by the local BMS **16**. Another of the local performance metrics may be associated with comfort provided by the local BMS **16**. Another of the local performance metrics may be associated with energy usage by the local BMS **16**. These are just examples.
(18) The controller **28** may be configured to aggregate like ones of the plurality of local performance metrics from the plurality of remote sites **14**, resulting in a plurality of aggregated performance metrics. For example, the local performance metrics associated with alarms from each of the remote sites may be rolled up or aggregated into one or more aggregated performance metrics associated with alarms. Aggregating may, for example, include one or more of averaging like ones of the plurality of local performance metrics from the plurality of remote sites **14** or summing like ones of the plurality of local performance metrics from the plurality of remote sites **14**. Alternatively, or in addition, aggregating may include computing a score based on like ones of the plurality of local performance metrics from the plurality of remote sites **14** and/or ranking like ones of the plurality of local performance metrics from the plurality of remote sites **14**. These are just examples.
(19) The controller **28** may display on the display **26** a plurality of panels, each panel associated with a different one of the plurality of local performance metrics. The controller **28** may also display in each panel the corresponding one of the plurality of aggregated performance metrics. In some cases, the controller **28** may display in each of the plurality of panels a ranking of one or more of the remote sites **14** by their corresponding local performance metric.
(20) In some cases, a first one of the plurality of panels that are displayed on the display **26** may be associated with a first local performance metric that is associated with alarms that are issued by the local BMS **16**. A second one of the plurality of panels that are displayed on the display **26** may be associated with a second local performance metric that is associated with comfort provided by the local BMS **16**. A third one of the plurality of panels that are displayed on the display **26** may be associated with a third local performance metric that is associated with energy usage by the local BMS **16**. These are just examples.
(21) In some instances, the controller **28** may be configured to process each of the plurality of local performance metrics of each of the plurality of remote sites **14** to identify those that do not meet a predefined criteria. The controller **28** may be configured to classify each of the plurality of local performance metrics of each of the plurality of remote sites **14** that do not meet the predefined criteria as needing attention. In some cases, the controller **28** may be configured to aggregate like ones of the plurality of local performance metrics from the plurality of remote sites **14** that are classified as needing attention, and display an indication of the aggregation of those needing attention on the corresponding one of the plurality of panels.
(22) In some cases, the controller **28** may be configured to display a map view adjacent the plurality of panels, wherein the map view displays a geographical location of at least some of the plurality of remote sites **14**. The controller **28** may allow a user to select a sub-set of the plurality of remote sites **14** on the map view, and in response, aggregate like ones of the plurality of local performance metrics from only the sub-set of the plurality of remote sites **14**, and display in each of the plurality of panels the corresponding one of the plurality of aggregated performance metrics for only the sub-set of the plurality of remote sites.
(23) The controller **28** may be configured to allow a user to select one of the plurality of remote sites **14** and, in response to selection of one of the plurality of remote sites **14**, display a site view that includes at least some of the local performance metrics associated with the particular selected remote site **14**. In some cases, the controller **28** is also configured to, in response to selection of one of the plurality of remote sites **14**, display performance indicators associated with one or more pieces of equipment **18**, **20** that are part of the local BMS **16** at the selected remote site and to allow a user to select one of the pieces of equipment **18**, **20** that are part of the local BMS **16** at the selected remote site **14**. In response to selection of one of the pieces of equipment **18**, **20** that are part of the local BMS **16** at the selected remote site **14**, the controller **28** is configured to display an equipment view that includes additional information associated with the operation

of the to select one of the pieces of equipment **18**, **20**. In some cases, the additional information associated with the operation of the selected one of the pieces of equipment may include one or more alarms issued by the selected one of the pieces of equipment **18**, **20**, sensor values associated with the operation of the select one of the pieces of equipment **18**, **20**, control signals associated with the operation of the select one of the pieces of equipment **18**, **20**, and/or a schedule associated with the select one of the pieces of equipment **18**, **20**.

(24) FIG. **3** is a flow diagram showing an illustrative method **30** that may be carried out by the multi-site BMS **12**. A plurality of local performance metrics associated with a local BMS of each of a plurality of remote sites **14** is determined based on operational data received from the local BMS **16** of each of the plurality of remote sites **14**, as indicated at block **32**. Like ones of the plurality of local performance metrics from the plurality of remote sites **14** are aggregated, resulting in a plurality of aggregated performance metrics, as indicated at block **34**. A plurality of panels are displayed on the display **26**, each panel being associated with a different one of the plurality of local performance metrics, as indicated at block **36**. The corresponding one of the plurality of aggregated performance metrics are displayed in each of the plurality of panels, as indicated at block **38**.

(25) As an example, a first one of the plurality of panels may be associated with a first local performance metric such as alarms that are issued by the local BMS **16**. A second one of the plurality of panels may be associated with a second local performance metric such as comfort provided by the local BMS **16**. A third one of the plurality of panels may be associated with a third local performance metric such as energy usage by the local BMS **16**. A user is allowed to select one of the plurality of remote sites **14**, as indicated at block **40**. In response to selection of one of the plurality of remote sites **14**, a site view is displayed that includes at least some of the local performance metrics associated with the particular selected remote site **14**, as indicated at block **42**.

(26) FIG. **4** is a flow diagram showing an illustrative method **50** that may be carried out by the multi-site BMS **12**. A plurality of local performance metrics associated with a local BMS of each of a plurality of remote sites **14** is determined based on operational data received from the local BMS **16** of each of the plurality of remote sites **14**, as indicated at block **32**. Like ones of the plurality of local performance metrics from the plurality of remote sites **14** are aggregated, resulting in a plurality of aggregated performance metrics, as indicated at block **34**. A plurality of panels are displayed on the display **26**, each panel being associated with a different one of the plurality of local performance metrics, as indicated at block **36**. The corresponding one of the plurality of aggregated performance metrics are displayed in each of the plurality of panels, as indicated at block **38**. A user is allowed to select one of the plurality of remote sites **14**, as indicated at block **40**.

(27) In response to selection of one of the plurality of remote sites **14**, performance indicators associated with one or more pieces of equipment **18**, **20** that are part of the local BMS **16** at the selected remote site **14** are displayed, as indicated at block **52**. A user is allowed to select one of the pieces of equipment **18**, **20** that are part of the local BMS **16** at the selected remote site **14**, as indicated at block **54**. In response to selection of one of the pieces of equipment **18**, **20** that are part of the local BMS **16** at the selected remote site **14**, an equipment view is displayed that includes additional information associated with the operation of the selected one of the pieces of equipment **18**, **20**, as indicated at block **56**. In some cases, the additional information associated with the operation of the selected one of the pieces of equipment **18**, **20** may include one or more of alarms issued by the selected one of the pieces of equipment **18**, **20**, sensor values associated with the operation of the select one of the pieces of equipment **18**, **20**, control signals associated with the operation of the select one of the pieces of equipment **18**, **20** and a schedule associated with the select one of the pieces of equipment **18**, **20**.

(28) FIG. **5** is a flow diagram showing an illustrative method **60** for monitoring a performance of a local BMS **16** at each of a plurality of remote sites **14**. A plurality of local performance metrics associated with a local BMS **16** of each of a plurality of remote sites **14** are determined based on operational data received from the local BMS **16** of each of the plurality of remote sites **14**, as indicated at block **62**. Like ones of the plurality of local performance metrics from the plurality of remote sites **14** are aggregated, resulting in a plurality of aggregated performance metrics, as indicated at block **64**. In some cases, aggregating includes one or more of averaging like ones of the plurality of local performance metrics from the plurality of remote sites **14** or summing like ones of the plurality of local performance metrics from the plurality of remote sites **14**. Aggregating may also include one or more of computing a score based on like ones of the plurality of local performance metrics from the plurality of remote sites **14** and ranking like ones of the plurality of local

performance metrics from the plurality of remote sites **14**.

(29) A plurality of panels are displayed on the display **26**, each panel associated with a different one of the plurality of local performance metrics, as indicated at block **66**. The corresponding one of the plurality of aggregated performance metrics are displayed in each of the plurality of panels, as indicated at block **68**. A ranking of one or more of the remote sites **14** by their corresponding local performance metric are displayed, at indicated at block **70**.

(30) FIGS. **6** through **11** are screen shots showing examples of some of the screens that may be generated by the multi-site BMS **12**. FIG. **6** shows a portfolio level dashboard **80**. In some cases, as illustrated, the portfolio level dashboard **80** includes a map **82** that shows a geographic area in which a number of remote sites **14** are located. As shown, the map **82** includes several icons **84** that each represent one or more remote sites **14**. For example, an icon **84***a* represents a total of four remote sites **14**, an icon **84***b* represents a total of three remote sites **14**, an icon **84***c* represents a single remote site **14**, an icon **84***d* represents a total of two remote sites **14** and an icon **84***e* represents a single remote site **14**. Each of the icons **84** may be selected in order to view additional information regarding the remote sites **14** that are represented by the particular icon **84**.

(31) In some cases, the icons **84** may be color coded. For example, a first color may represent alarms, a second color may represent comfort and a third color may represent energy. In some instances, different colors may be used to represent varying degrees of seriousness. For example, red may be used to indicate that there is a serious alarm at one of the remote sites **14** while yellow may be used to indicate a less serious alarm at one of the remote sites **14**. Various colors may be used to indicate how many problems are detected at a particular remote site **14**, for example.

(32) The portfolio level dashboard **80** includes a number of panels. As illustrated, the portfolio level dashboard **80** includes an Alarm panel **86**, a Comfort panel **88** and an Energy panel **90**. In some instances, a user may determine that they are not interested in comfort, for example, and the controller **28** may be configured to no longer display the Comfort panel **88**. This is just an example. The Alarm panel **86** may include a Reported Alarms icon **92** that shows how many alarms have been reported, an Active Alarms icon **94** that shows how many alarms are currently active and an Alarms Listing icon **96** that provides a listing of how many high alarms, how many medium alarms and how many low alarms are present. The Alarm panel **86** also includes a listing **98** of the site rankings of the remote sites **14** reporting alarms. The listing **98** may be sorted, if desired, to reveal superior performing sites and/or underperforming sites based on total number of alarms, number of unresolved alarms, number of serious alarms, frequency of alarms, average time taken to resolve an alarm, and or any other suitable criteria.

(33) In the example shown, the Comfort panel **88** includes an overall Comfort Score icon **100** that provides a visual indication of an overall comfort score of the remote sites. The Comfort panel **88** also includes a listing **102** of the parameters being used to determine the overall comfort score. As shown, the overall comfort score is based at least in part upon an average temperature score, a humidity score and a carbon dioxide (CO2) score. The Comfort panel **88** also includes a listing **104** of particular sites contributing to the overall comfort score. The listing **104** may be sorted, if desired, to reveal superior performing sites and/or underperforming sites. For example, each remote site may have a computed local comfort score based on the performance of the local BMS, and the listing **104** may be sorted by the local comfort score of each site. This is just one example.

(34) In the example shown, the Energy panel **90** includes an Excess Use icon **106** that shows how many sites are reporting excessive energy usage, a Factory Default Schedule icon **108** that shows how many sites are using a factory default schedule and a Manual Override icon **110** that shows how many sites are operating under a manual override. The Energy panel **90** also includes a listing **112** that shows the sites contributing to the Excess Use icon **106**, the Factory Default Schedule icon **108** and the Manual Override icon **110**. The listing **112** may be sorted, if desired, to reveal superior performing sites and/or underperforming sites.

(35) In the example shown, each of the Alarm panel **86**, the Comfort panel **88** and the Energy panel **90** include a Current button **114** and Trend button **116**. The Current button **114** may be selected to display current information (as is shown in FIG. **6**). The Trend button **116** may be selected to display historical data including historical trends. In some cases, historical data may be shown in graphical form within the appropriate panel such as the Alarm panel **86**, the Comfort panel **88** and the Energy panel **90**.

(36) In some cases, it is possible to toggle between the portfolio level dashboard **80** shown in FIG. **6** with a map view and a portfolio dashboard **120** with a list view. An example list view is shown in FIG. **7**. The portfolio level dashboards **80**, **120** include a map view icon **122** and a list view icon **124**. It can be seen that

in FIG. **6**, the map view icon **122** has been selected while in FIG. **7**, the list view icon **124** has been selected. The portfolio level dashboard **120** in list view includes a row **126** that provides information as to the number of sites, how many sites are offline, how many are currently in alarm, how many are currently using too much energy, and the like.

(37) The portfolio level dashboard **120** in list view includes a Sites column **128**, an Alarm total column **130**, an Active High Alarm column **132**, a Comfort Score column **134**, a Temperature Score column **136**, a Humidity Score column **138**, a CO2 Score column **140**, an Excess Use column **142**, a Factory Default Schedule column **144**, a Manual Override column **146** and an Override Duration column **148**. It will be appreciated that much of the information provided in the portfolio level dashboard **80** in map view is also shown in the portfolio level dashboard **120** in list view. A point of interest in the Site column **128** is that sites are organized in a hierarchal manner, with individual components listed under their corresponding header. For example, the header LDS 7350 High River has been expanded to reveal Chapel, F3 RS RM Bishop, and so on. In the example shown, Chapel, F3 RS RM Bishop are each individual pieces of equipment (e.g. a rooftop unit) at the LDS 7350 High River site.

(38) FIG. **8** shows a site level dashboard **160** that shows an equipment list view while FIG. **9** shows a site level dashboard **190** that shows a device list view. The site level dashboards **160**, **190** may be reached by selecting the site on the portfolio level dashboard. In some cases, a user may enter a search query into the multi-site BMS to identify a desired remote site, and then select the site to reach the desired site level dashboards **160**, **190**. These are just example.

(39) In the example shown, once the site level dashboard **160**, **190** is reached, a user is able to select between the equipment list view and the device list view by toggling either an equipment view icon **162** or a device list view icon **164**. The site level dashboard **160** allows a user to see all of the equipment at a particular site in a single list that allows the user to switch between different equipment types such as but not limited to RTU (roof top units), VRF (variable refrigerant flow units) and AHU (air handling units). As illustrated, roof top units have been selected. The site level dashboard **160** includes a Name column **166**, a Current Status column **168**, an Active High Alarm column **170**, a Current Temperature column **172**, an Effective Setpoint column **174**, a Humidity column **176**, an Excess Runtime column **178**, a Manual Override Duration column **180** and a Current Schedule column **182**.

(40) The site level dashboard **190** with the device list view icon **164** selected shows information for devices such as sensors, lighting and the like. FIG. **9** shows a sensor summary list. The site level dashboard **190** includes a Name column **192**, a Zone column **194**, a Current Status column **196**, an Active Alarm column **198**, a Current Value column **200**, an RSSI column **202**, a Battery column **204** and a Firmware Update column **206**.

(41) FIG. **10** provides an example of an equipment level dashboard **220** that allows a user to monitor, command and control the current status, parameter values and/or control signals for a selected piece of equipment. In some cases, a multiple objects trend view allows visualization of each parameter over time. This can provide for improved user interaction and interpretation for better trouble shooting. Scheduling of the equipment can also be seen. In some cases, the equipment level dashboard **220** may be configured to have a generic design that can adapt to any of a variety of different types of equipment without requiring additional configuration. The equipment level dashboard **220** may be reached by selecting the appropriate piece of equipment on the site level dashboard **160**, for example.

(42) As noted, in some cases, the user may be able to specify which panels are displayed on the portfolio level dashboard **80**. FIG. **11** provides a screen **240** that may be used to specify whether the Comfort panel **88** is displayed. Similar screens may be displayed (not shown) to specify whether the Alarms panel **86** and/or the Energy panel **90** will be displayed. The screen **240** includes a slider **242** that may be switched between enable and disable. If enabled, the Comfort panel **88** will be displayed. If disabled, the Comfort panel **88** will not be displayed. In some cases, if the Comfort panel **88** is not displayed, the other panels such as the Alarms panel **86** and the Energy panel **90** may be displayed over a larger portion of the screen. The screen **240** also includes a section **244** that allows the user to select alarm limits for comfort. The screen **240** also includes a section **246** that allows the user to select which particular parameters will be included in calculating overall scores.

(43) Having thus described several illustrative embodiments of the present disclosure, those of skill in the art will readily appreciate that yet other embodiments may be made and used within the scope of the claims hereto attached. It will be understood, however, that this disclosure is, in many respects, only illustrative. Changes may be made in details, particularly in matters of shape, size, arrangement of parts, and exclusion

and order of steps, without exceeding the scope of the disclosure. The disclosure's scope is, of course, defined in the language in which the appended claims are expressed.

Claims

1. A multi-site Building Management System (BMS) for monitoring a performance of and controlling a local BMS at each of a plurality of remote sites, wherein each of the plurality of remote sites includes a local BMS, the multi-site BMS comprising: a port for receiving operational data as the operational data is coming in from the local BMS of each of the plurality of remote sites; a display; a controller operatively coupled to the display and the port, the controller configured to: determine a plurality of local performance metrics associated with the local BMS of each of the plurality of remote sites based on the operational data, wherein the plurality of local performance metrics include different local performance metric types including: a first performance metric type that is associated with alarms issued by the corresponding local BMS; a second performance metric type that is associated with comfort provided by the corresponding the local BMS; and a third performance metric type that is associated with energy usage of the corresponding the local BMS; process each of the plurality of local performance metrics of each of the plurality of remote sites to identify one or more non-compliant local performance metrics that do not meet a corresponding predefined criteria; identify like ones of the plurality of local performance metrics of like performance metric type from the plurality of remote sites; aggregate the like ones of the plurality of local performance metrics of like performance metric type from the plurality of remote sites, resulting in at least one aggregated performance metric for each performance metric type; concurrently display on the display a plurality of panels, wherein each panel is defined by a visual boundary that extends around a perimeter of the respective panel, and each panel is associated with a different one of the plurality of local performance metric types; display in each of the plurality of panels at least one of the aggregated performance metrics associated with the corresponding performance metric type; display a map view concurrently with the plurality of panels, wherein the map view displays a geographical location of at least some of the plurality of remote sites; receive via the map view a user selection of one of the plurality of remote sites via a user interface device associated with the BMS, and in response, update the display to display a site view on the display for the selected one of the plurality of remote sites, wherein the site view for the selected one of the plurality of remote sites displays at least some of the local performance metrics associated with the particular selected remote site and one or more pieces of equipment of the local BMS of the particular selected remote site; receive via the site view of the selected one of the plurality of remote sites a selection of one or more of the pieces of equipment of the local BMS of the particular selected remote site via the user interface device associated with the BMS, and in response, update the display to display an equipment view on the display, wherein the equipment view displays additional information associated with an operation of the selected one or more pieces of equipment of the local BMS of the particular selected remote site; determine and download one or more control commands to the local BMS of the particular selected remote site by to control one or more pieces of equipment of the corresponding local BMS to improve one or more of the non-compliant local performance metrics; and the corresponding local BMS executing the one or more downloaded control commands causing the corresponding local BMS to control one or more pieces of equipment of the corresponding local BMS to improve one or more of the non-compliant local performance metrics.
2. The multi-site Building Management System (BMS) of claim 1, wherein the controller is configured to allow a user selection of a sub-set of the plurality of remote sites via the map view, and in response, aggregate like ones of the plurality of local performance metrics of like performance metric type from only the sub-set of the plurality of remote sites, and update the display to display in each of the plurality of panels the corresponding one of the plurality of aggregated performance metrics for only the sub-set of the plurality of remote sites.
3. The multi-site Building Management System (BMS) of claim 2, wherein the sub-set of the plurality of remote sites includes two or more of the plurality of remote sites.
4. The multi-site Building Management System (BMS) of claim 1, wherein the additional information associated with the operation of the selected one of the pieces of equipment includes one or more of: alarms issued by the selected one of the pieces of equipment; sensor values associated with the operation of the select one of the pieces of equipment; control signals associated with the operation of the select one of the pieces of equipment; and a schedule associated with the select one of the pieces of equipment.

5. The multi-site Building Management System (BMS) of claim 1, wherein the controller is configured to classify each of the plurality of local performance metrics of each of the plurality of remote sites that do not meet the corresponding predefined criteria as needing attention.
6. The multi-site Building Management System (BMS) of claim 5, wherein the controller is configured to aggregate like ones of the plurality of local performance metrics of like performance metric type from the plurality of remote sites that are classified as needing attention, and display an indication of the aggregation of those needing attention on the corresponding one of the plurality of panels.
7. The multi-site Building Management System (BMS) of claim 1, wherein aggregating comprises one or more of: averaging like ones of the plurality of local performance metrics of like performance metric type from the plurality of remote sites; summing like ones of the plurality of local performance metrics of like performance metric type from the plurality of remote sites; and computing a score based on like ones of the plurality of local performance metrics of like performance metric type from the plurality of remote sites.
8. The multi-site Building Management System (BMS) of claim 1, wherein: a first one of the plurality of panels is associated with the first local performance metric type; and a second one of the plurality of panels is associated with the second local performance metric type.
9. A non-transient computer readable medium storing thereon instructions that when executed by a processor cause the processor to: determine a plurality of local performance metrics associated with a local BMS of each of a plurality of remote sites based on operational data, wherein the plurality of local performance metrics include different local performance metric types including: a first performance metric type that is associated with alarms issued by the corresponding local BMS; a second performance metric type that is associated with comfort provided by the corresponding the local BMS; and a third performance metric type that is associated with energy usage of the corresponding the local BMS; process each of the plurality of local performance metrics of each of the plurality of remote sites to identify one or more non-compliant local performance metrics that do not meet a corresponding predefined criteria; identify like ones of the plurality of local performance metrics of like performance metric type from the plurality of remote sites; aggregate like ones of the plurality of local performance metrics of like performance metric type from the plurality of remote sites, resulting in at least one aggregated performance metric for each performance metric type; concurrently display on a display a plurality of panels, wherein each panel is defined by a visual boundary that extends around a perimeter of the respective panel, and each panel is associated with a different one of the plurality of local performance metric types; display in each of the plurality of panels at least one of the aggregated performance metrics associated with the corresponding performance metric type; display a map view concurrently with the plurality of panels, wherein the map view displays a geographical location of at least some of the plurality of remote sites; allow via the map view a user selection of one of the plurality of remote sites via a user interface device; in response to selection of one of the plurality of remote sites via the map view, update the display to display a site view for the selected one of the plurality of remote sites that displays at least some of the local performance metrics associated with the particular selected remote site and one or more pieces of equipment of the local BMS of the particular selected remote site; display performance indicators associated with one or more of the pieces of equipment of the local BMS of the particular selected remote site; receive via the site view of the selected one of the plurality of remote sites a selection of one or more of the pieces of equipment of the local BMS of the particular selected remote site via the user interface device, and in response, update the display to display an equipment view that displays additional information associated with an operation of the selected one of the pieces of equipment of the local BMS of the particular selected remote site; determine and download one or more control commands to the local BMS of the particular selected remote site to control one or more pieces of equipment of the corresponding local BMS to improve one or more of the non-compliant local performance metrics; and the corresponding local BMS executing the one or more downloaded control commands causing the corresponding local BMS to control one or more pieces of equipment of the corresponding local BMS to improve one or more of the non-compliant local performance metrics.
10. The non-transient computer readable medium of claim 9, wherein the additional information associated with the operation of the selected one of the pieces of equipment of the local BMS of the particular selected remote site includes one or more of: alarms issued by the selected one of the pieces of equipment; sensor values associated with the operation of the select one of the pieces of equipment; control signals associated with the operation of the select one of the pieces of equipment; and a schedule associated with the select one of the pieces of equipment.
11. The non-transient computer readable medium of claim 9, wherein aggregating comprises one or more of:

averaging like ones of the plurality of local performance metrics of like performance metric type from the plurality of remote sites; summing like ones of the plurality of local performance metrics of like performance metric type from the plurality of remote sites; and computing a score based on like ones of the plurality of local performance metrics of like performance metric type from the plurality of remote sites.

12. A method for monitoring a performance of a local BMS at each of a plurality of remote sites, wherein each of the plurality of remote sites includes a local BMS that includes one or more pieces of equipment, the method comprising: determining a plurality of local performance metrics associated with a local BMS of each of a plurality of remote sites based on operational data, wherein the plurality of local performance metrics include different local performance metric types including: a first performance metric type that is associated with alarms issued by the corresponding local BMS; a second performance metric type that is associated with comfort provided by the corresponding the local BMS; and a third performance metric type that is associated with energy usage of the corresponding the local BMS; processing each of the plurality of local performance metrics of each of the plurality of remote sites to identify one or more non-compliant local performance metrics that do not meet a corresponding predefined criteria; identifying like ones of the plurality of local performance metrics of like performance metric type from the plurality of remote sites; aggregating the like ones of the plurality of local performance metrics of like performance metric type from the plurality of remote sites, resulting in at least one aggregated performance metric for each performance metric type; concurrently displaying on a display a plurality of panels, wherein each panel is defined by a visual boundary that extends around a perimeter of the respective panel, and each panel is associated with a different one of the plurality of local performance metric types; displaying in each of the plurality of panels at least one of the aggregated performance metrics associated with the corresponding performance metric type; displaying a map view concurrently with the plurality of panels, wherein the map view displays a geographical location of at least some of the plurality of remote sites; receiving via the map view a user selection of one of the plurality of remote sites via a user interface device, and in response, updating the display to display a site view on the display for the selected one of the plurality of remote sites, wherein the site view for the selected one of the plurality of remote sites displays at least some of the local performance metrics associated with the particular selected remote site and one or more pieces of equipment of the local BMS of the particular selected remote site; receiving via the site view of the selected one of the plurality of remote sites a selection of one or more of the pieces of equipment of the local BMS of the particular selected remote site via the user interface device, and in response, updating the display to display an equipment view on the display, wherein the equipment view displays additional information associated with an operation of the selected one or more pieces of equipment of the local BMS of the particular selected remote site; and determining and downloading one or more control commands to the local BMS of the particular selected remote site to control one or more pieces of equipment of the corresponding local BMS to improve one or more of the non-compliant local performance metrics; and the corresponding local BMS executing the one or more downloaded control commands causing the corresponding local BMS to control one or more pieces of equipment of the corresponding local BMS to improve one or more of the non-compliant local performance metrics.

13. The method of claim 12, comprising: allowing a user selection of a sub-set of the plurality of remote sites via the map view, and in response, aggregate like ones of the plurality of local performance metrics of like performance metric type from only the sub-set of the plurality of remote sites, and updating the display to display in each of the plurality of panels the corresponding one of the plurality of aggregated performance metrics for only the sub-set of the plurality of remote sites.

14. The method of claim 12, wherein the additional information includes one or more of: alarms issued by the selected one of the pieces of equipment; sensor values associated with the operation of the select one of the pieces of equipment; control signals associated with the operation of the select one of the pieces of equipment; and a schedule associated with the select one of the pieces of equipment.